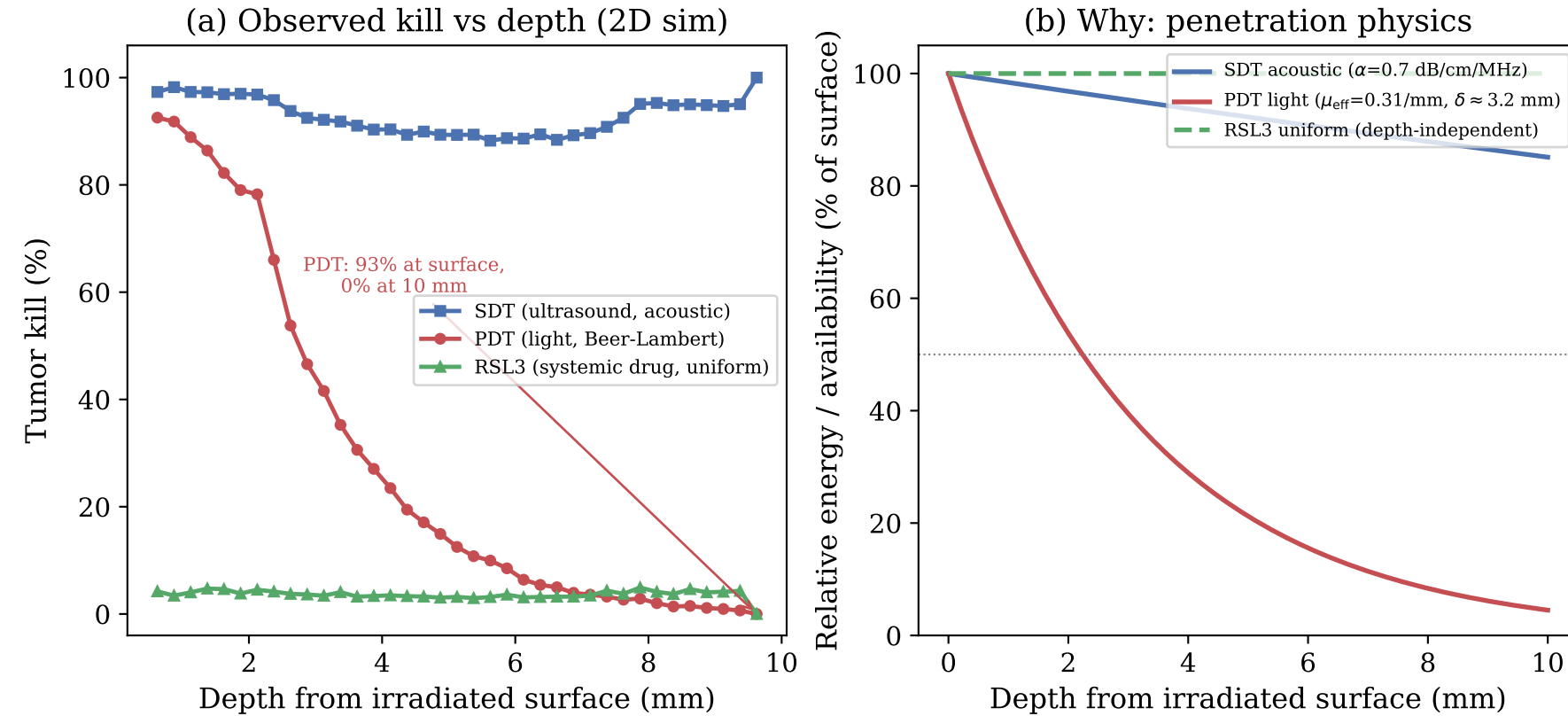


Penetration sets modality reach: light is millimeters, ultrasound is centimeters (2D model)



2D sim-spatial, 1 cm tissue. Depth profiles follow well-measured physics (Beer-Lambert optics, acoustic attenuation; high confidence). RSL3 reaches every depth but kills little, a biochemical limit, not a penetration one. Absolute kill % rests on uncalibrated biochemistry, so read the profile shape, not the magnitudes. SDT is modeled as O_2 -independent, an optimistic upper bound (Section 7.1).